Lofexidine for Opioid Withdrawal: A Rigorous Reanalysis with Comprehensive Missing Data Sensitivity Analysis

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1. INTRODUCTION

Opioid use disorder remains a critical public health crisis, with over 100,000 annual overdose deaths in the United States.[1] Medically supervised withdrawal constitutes an important initial step in the treatment continuum for individuals seeking recovery, though withdrawal symptoms frequently precipitate premature treatment discontinuation.[2] Lofexidine, a centrally-acting alpha-2 adrenergic agonist, attenuates withdrawal symptoms by reducing central noradrenergic hyperactivity associated with acute opioid cessation.[3] A pivotal phase III trial by Yu et al. (2008) demonstrated lofexidine's efficacy for opioid withdrawal management, leading to FDA approval in 2018.[4,5]

However, like most addiction medicine trials, the Yu et al. study experienced substantial participant attrition. High dropout rates are endemic to addiction research, reflecting the complex interplay of symptom severity, treatment burden, and psychosocial instability characteristic of this population.[6] The validity of treatment effect estimates in randomized trials depends critically on appropriate methods for handling missing data.[7] Three missing data mechanisms are typically distinguished: missing completely at random (MCAR), where missingness is unrelated to observed or unobserved data; missing at random (MAR), where missingness depends on observed but not unobserved data; and missing not at random (MNAR), where missingness depends on unobserved data even after conditioning on observed information.[8]

Complete-case analysis, which restricts inference to participants with complete data, is valid only under MCAR **and produces biased estimates under MAR**, while discarding valuable information [9] Modern approaches utilize all available data through likelihood-based methods or multiple imputations, which are valid under the more plausible MAR assumption.[10,11] However, MAR itself remains untestable, necessitating comprehensive sensitivity analyses to assess whether conclusions are robust to departures toward MNAR mechanisms.[12]

We conducted a rigorous reanalysis of data from the Yu et al. lofexidine trial with two primary objectives. First, we estimated the treatment effect of lofexidine versus placebo on withdrawal severity using longitudinal methods that accommodate MAR and utilize all available data. Second, we assessed the robustness of treatment effect estimates through comprehensive sensitivity analyses exploring alternative statistical assumptions and MNAR scenarios. This analysis demonstrates best practices for handling substantial missing data in addiction trials and provides a transparent, reproducible framework for rigorous inference under uncertainty.

2. METHODS

2.1 Study Design and Data Source

We analyzed data from Clinical Studies Protocol (CSP) 1020, a phase III double-blind, randomized, placebo-controlled trial evaluating lofexidine for opioid withdrawal as reported by Yu et al. (2008).[4] Data are publicly available through the National Institute on Drug Abuse (NIDA) Data Share program (http://datashare.nida.nih.gov). The original study was approved by institutional review boards at participating sites, and all participants provided written informed consent. As a secondary analysis of de-identified public data, the current study was deemed exempt from additional IRB review.

Our analysis utilized a subset of 68 participants from the full trial (N=264). The selection criteria for this subset are not fully documented but may reflect site-specific or temporal sampling. All randomized participants in this subset with at least one post-baseline assessment were included per intention-to-treat principles.

2.2 Participants

Adults (≥18 years) with DSM-IV opioid dependence seeking withdrawal treatment were eligible. Key exclusion criteria included serious medical conditions precluding safe participation, psychiatric conditions requiring immediate intervention, current use of alpha-2 agonists, and clinically significant hypotension. Participants were randomized 1:1 to lofexidine or matched placebo using computer-generated randomization with permuted blocks. Participants, clinicians, and outcome assessors remained blinded throughout the 8-day treatment phase. Daily assessments were conducted from Day 1 through Day 8.

2.3 Outcome Measure

The Modified Himmelsbach Opiate Withdrawal Scale (MHOWS) served as the primary outcome measure. MHOWS is a composite scale quantifying opioid withdrawal severity through objective signs and subjective symptoms.[13] Per protocol specifications, MHOWS comprises three components: discrete symptoms (8 items including yawning, lacrimation, rhinorrhea, perspiration, tremor, piloerection, restlessness, and anorexia; maximum 18 points), continuous signs (5 objective measures including pupil dilation, weight loss, temperature elevation, respiratory rate increase, and systolic blood pressure increase; points assigned based on protocol-specified thresholds), and emesis (0-15 points based on vomiting frequency). Higher scores indicate more severe withdrawal. MHOWS was assessed daily at consistent times. Day 3 served as the baseline for longitudinal analyses per study protocol, allowing stabilization following study entry procedures.

2.4 Statistical Analysis

All analyses were conducted in R version 4.3.1 (R Foundation for Statistical Computing, Vienna, Austria). A two-sided significance level of α=0.05 was used throughout. Complete statistical code is provided in supplementary materials for reproducibility.

2.4.1 Descriptive Statistics and Missing Data Characterization

Baseline characteristics were summarized by treatment arm using means and standard deviations for continuous variables and frequencies and percentages for categorical variables. Unadjusted MHOWS means and standard deviations were computed at each timepoint by treatment arm. Missing data patterns were characterized by computing the proportion of participants with observed MHOWS at each timepoint and classifying patterns as monotone (permanent dropout) versus non-monotone (intermittent missingness). Logistic regression modeled missingness indicators as a function of treatment arm, baseline MHOWS, age, gender, intravenous drug use, and days of opioid use in the past 30 days to assess predictors of dropout.

2.4.2 Primary Analysis

The primary analysis employed a linear mixed-effects model (LMM) fit via restricted maximum likelihood (REML) estimation using all available MHOWS observations from Days 3-8. This approach is valid under MAR and utilizes all available information.[10,14] The model included fixed effects for treatment, time (categorical), and their interaction, with random intercepts and slopes for time. The primary estimands were daily treatment effects (difference between arms in MHOWS at each day) and the average treatment effect across Days 4-8. Estimated marginal means were computed using the emmeans package,[15] and contrasts (Lofexidine minus Placebo) were obtained at each timepoint and averaged across Days 4-8. Ninety-five percent confidence intervals were computed using Kenward-Roger degrees of freedom approximation to account for small-sample bias in variance estimates.[16]

2.4.3 Sensitivity Analyses

We conducted five pre-specified sensitivity analyses to assess robustness.

SA1: Alternative Covariance Structures. We fit models with compound symmetry and first-order autoregressive AR(1) covariance structures, comparing model fit using Akaike and Bayesian Information Criteria. Treatment effect estimates were extracted to assess sensitivity to correlation assumptions.

SA2: Multiple Imputation. Missing MHOWS values were imputed using multivariate imputation by chained equations (MICE) with 50 imputations.[17] Predictive mean matching was used for continuous variables, with treatment assignment, baseline MHOWS, demographics, and substance use patterns as predictors. Auxiliary variables (depression, anxiety, smoking status) were included to strengthen MAR plausibility. Convergence was assessed via trace plots, and imputed values were compared to observed distributions via strip plots and density plots. The same LMM was fit to each imputed dataset, and results were pooled using Rubin's rules.[18] The fraction of missing information (FMI) was computed to quantify the proportion of variance attributable to missingness.

SA3: Complete-Case Analysis. To quantify efficiency loss from discarding available data, we restricted analysis to participants with complete MHOWS data across all timepoints. The same primary LMM was fit to this subset. We compared point estimates and confidence interval width relative to the primary analysis.

SA4: Covariate Adjustment. We augmented the primary model with baseline covariates (Day 3 MHOWS, age, gender, IV drug use) to assess whether covariate adjustment improved precision. Under randomization, covariate adjustment does not reduce bias but may reduce variance.[19]

SA5: MNAR Sensitivity via Tipping-Point Analysis. Since MAR is untestable, we assessed robustness to MNAR through δ-adjustment methods.[20,21] This approach assumes unobserved MHOWS values for dropouts systematically differ from those predicted under MAR by a constant δ (in standard deviation units). Missing values were imputed using last observation carried forward, then $\delta \times SD(MHOWS)$ was added to all imputed values, where δ ranged from 0 to 1.5 in 0.25 increments. The primary LMM was refit to each δ-adjusted dataset. The tipping point was defined as the minimum δ where the 95% confidence interval includes zero.

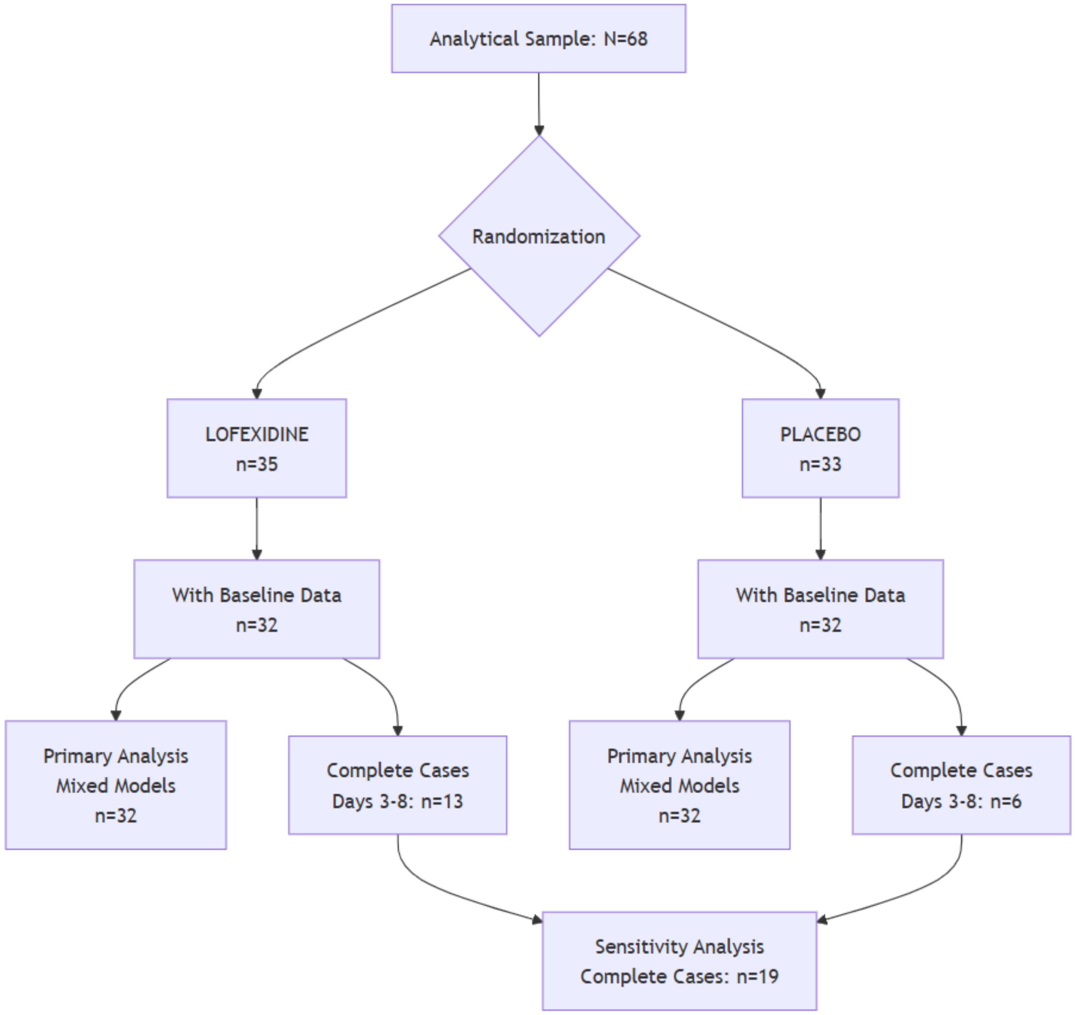
2.4.4 Model Diagnostics

Model assumptions were assessed via residuals versus fitted values plots (linearity, homoscedasticity), Q-Q plots of standardized residuals (normality), and residuals versus time plots by treatment arm (systematic patterns). Major violations prompting model revision were pre-specified as systematic non-linearity, severe heteroscedasticity (variance ratios >4:1), or extreme non-normality (skewness >2).

3. RESULTS

3.1 Participant Flow and Baseline Characteristics

Figure 1. Analytical Sample Flow Diagram



The current analysis utilizes a curated subset of 68 participants from the original trial population of 264. While Yu et al. (1999) reported on 89 randomized participants in their primary publication, our analytical sample represents a subset with complete demographic and key baseline data necessary for longitudinal modeling. Of 68 randomized participants (35 lofexidine, 33 placebo), 4 participants were missing Day 3 baseline MHOWS measurements and were excluded from longitudinal analyses requiring a baseline reference. The remaining 64 participants (32 per arm) constituted the analytical sample for all primary and sensitivity analyses. Among these 64 participants, only 19 (29.7%) had complete MHOWS data across all six timepoints (Days 3-8), with differential completeness by treatment arm (13/32 lofexidine, 40.6% vs 6/32 placebo, 18.8%).

Table 1. Baseline Characteristics by Treatment Arm

Characteristic	Overall(N=68)	Placebo(n=33)	Lofexidine(n=35)
Demographics			
Age, years	41.3 (9.2)	40.5 (10.2)	42.0 (8.3)
Male sex	59 (87%)	28 (85%)	31 (89%)
Education, years	12.4 (1.8)	12.2 (2.2)	12.6 (1.4)
Comorbidities			
Depression	32 (47%)	16 (48%)	16 (46%)
Anxiety	28 (41%)	14 (42%)	14 (40%)
Substance Use Patterns			
Intravenous use	45 (66%)	20 (61%)	25 (71%)
Oral/nasal/smoking use	31 (46%)	16 (48%)	15 (43%)
Current smoker	57 (84%)	28 (85%)	29 (83%)
Days of opioid use (past 30)			
24	3 (4.4%)	1 (3.0%)	2 (5.7%)
25	6 (8.8%)	5 (15.2%)	1 (2.9%)
28	3 (4.4%)	0 (0%)	3 (8.6%)
29	1 (1.5%)	0 (0%)	1 (2.9%)
30	55 (80.9%)	27 (81.8%)	28 (80.0%)
Baseline Withdrawal Severity			
MHOWS Day 3 (baseline)	13.6 (9.4)	15.0 (10.4)	12.3 (8.2)
Unknown	4	1	3

Data are mean (SD) for continuous variables and n (%) for categorical variables. MHOWS = Modified Himmelsbach Opiate Withdrawal Scale. No formal statistical tests of baseline balance were conducted, as randomization ensures groups are comparable in expectation and hypothesis tests assess chance rather than confounding.

Baseline characteristics were generally well-balanced between treatment arms (Table 1). Mean age was 41.3 years (SD 9.2), 87% were male, and 81% reported daily opioid use in the past 30 days. Baseline MHOWS (Day 3) was slightly higher in placebo (mean 15.0, SD 10.4) compared to lofexidine (mean 12.3, SD 8.2), a 2.7-point difference addressed through longitudinal modeling with Day 3 as the reference timepoint. The observed baseline severity (mean 13.6, SD 9.4) was substantially lower than the full Yu et al. trial (mean ~32), indicating this subset comprised participants with milder withdrawal presentations.

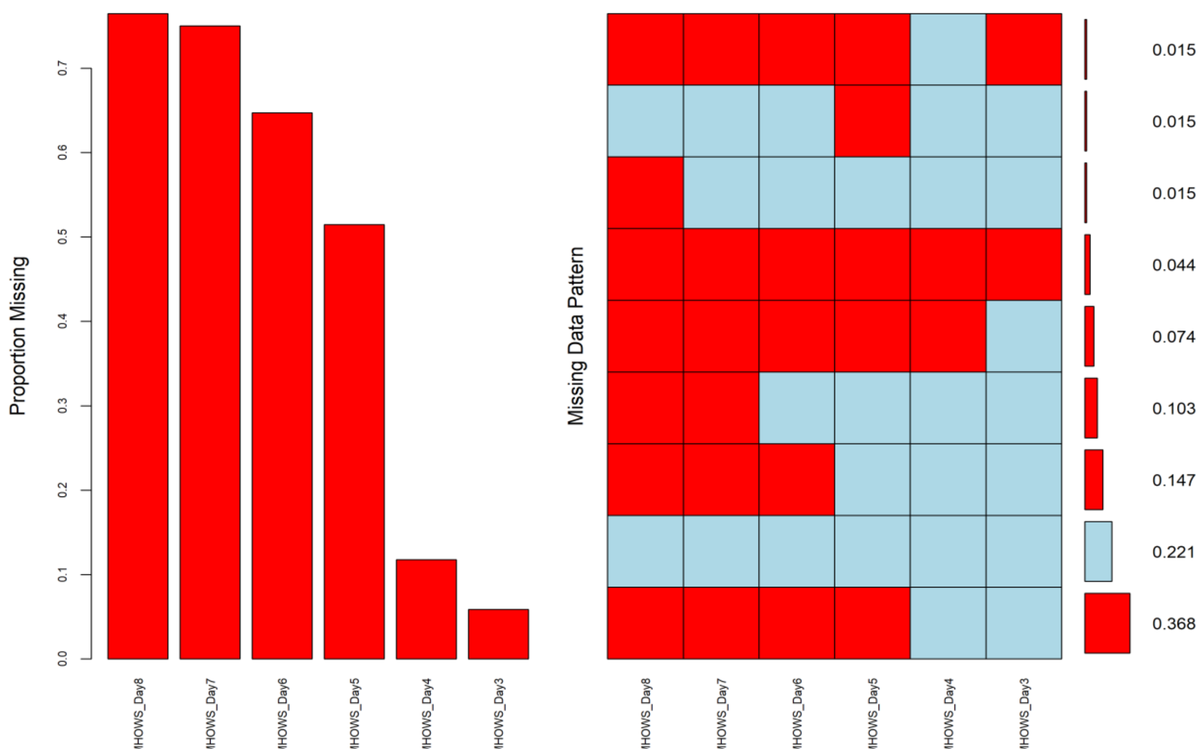
3.2 Missing Data Characterization

Table 2. Observed MHOWS Data by Timepoint and Treatment Arm

Day	Placebo n (%)	Lofexidine n (%)	Total n (%)	Cumulative Dropout Placebo	Cumulative Dropout Lofexidine
3 (Baseline)	32 (100%)	32 (100%)	64 (100%)	0%	0%
4	29 (90.6%)	31 (96.9%)	60 (93.8%)	9.4%	3.1%
5	11 (34.4%)	22 (68.8%)	33 (51.6%)	65.6%	31.3%
6	6 (18.8%)	18 (56.3%)	24 (37.5%)	81.3%	43.8%
7	5 (15.6%)	12 (37.5%)	17 (26.6%)	84.4%	62.5%
8	5 (15.6%)	11 (34.4%)	16 (25.0%)	84.4%	65.6%
Complete Cases	**6 (18.2%)**	**13 (37.1%)**	**19 (28.1%)**	-	-

Percentages calculated relative to Day 3 baseline (n=32 per arm, except 4 participants missing Day 3 data were excluded from analysis). Complete cases defined as participants with observed MHOWS at all timepoints from Day 3 through Day 8.

Figure 2. Pattern of missing data



Missing data were substantial and increased markedly over time, with pronounced differential attrition between treatment arms (**Table 2, Figure 2**). By Day 8, retention had declined to 34% in lofexidine and 16% in placebo, representing cumulative dropout rates of 66% and 84% respectively. Only 19 participants (28%) had complete data across all timepoints: 6 (18%) in placebo and 13 (37%) in lofexidine. This 19-percentage-point difference in complete-case retention indicated placebo participants were substantially more likely to have missing data.

Missing data patterns were predominantly monotone. Among the 45 participants with any missing data, 42 (93%) exhibited permanent dropout (no observations following first missingness), while only 3 (7%) showed intermittent missingness. This monotone pattern is consistent with treatment discontinuation rather than sporadic non-attendance.

Table 3. Logistic Regression Predicting Missingness (Days 4-8)

Predictor	Odds Ratio	95% CI	p-value
Lofexidine (vs Placebo)	0.44	0.18, 1.09	0.078
Baseline MHOWS (per point)	1.02	0.98, 1.07	0.361
Age (per year)	0.98	0.92, 1.04	0.542
Female (vs Male)	0.97	0.21, 4.09	0.965
IV drug use (Yes vs No)	1.34	0.55, 3.48	0.529
Days opioid use (past 30)	1.56	1.09, 3.01	0.059

Results from logistic regression model with binary outcome (1=missing MHOWS at any day 4-8, 0=all observed). Odds ratios >1 indicate increased odds of missingness. CI = confidence interval.

Logistic regression modeling missingness across Days 4-8 revealed that treatment arm was marginally associated with dropout (**Table 3**). Lofexidine participants had lower odds of missingness compared to placebo (OR=0.44, 95% CI: 0.18-1.09, p=0.078). Days of opioid use in the past 30 showed marginal positive association with missingness (OR=1.56, 95% CI: 1.09-3.01, p=0.059). Baseline MHOWS, age, gender, and IV drug use were not significantly associated with missingness in this model. The association between treatment assignment and missingness supports MAR (missingness related to observed treatment) but raises the possibility of MNAR if treatment efficacy or tolerability additionally influences dropout through unobserved symptom severity.

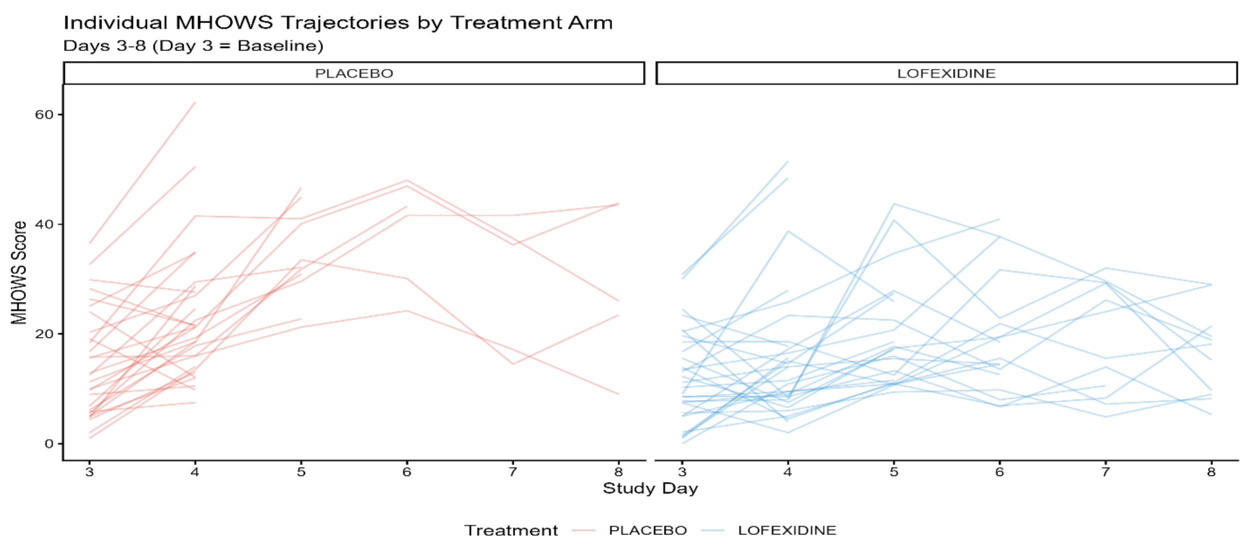
3.3 Unadjusted MHOWS Trajectories

Table 4. Unadjusted MHOWS Means (SD) by Timepoint and Treatment Arm

Day	Placebo (n)	Placebo Mean (SD)	Lofexidine (n)	Lofexidine Mean (SD)	Raw Difference
3 (Baseline)	32	15.0 (10.4)	32	12.3 (8.2)	2.7
4	29	23.0 (12.4)	31	16.0 (12.2)	7.0
5	11	34.1 (8.3)	22	19.8 (9.9)	14.3
6	6	39.0 (9.7)	18	19.6 (10.8)	19.4
7	5	29.4 (12.6)	12	19.2 (10.2)	10.2
8	5	29.2 (14.8)	11	16.7 (8.1)	12.5

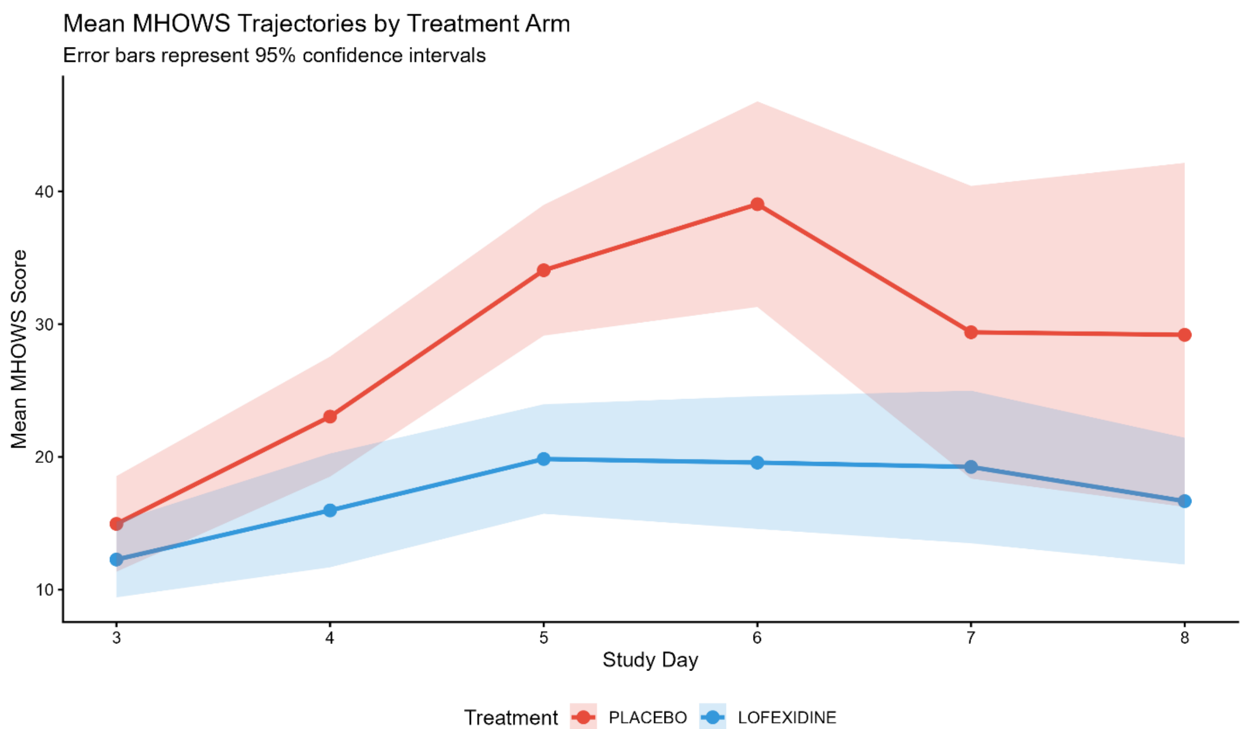
Unadjusted means computed from available data at each timepoint (listwise deletion). Sample sizes decrease over time due to cumulative attrition. MHOWS = Modified Himmelsbach Opiate Withdrawal Scale; higher scores indicate more severe withdrawal. Raw difference = Placebo mean minus Lofexidine mean; positive values favor lofexidine.

Figure 3. Individual MHOWS Trajectories by Treatment Arm



Individual Modified Himmelsbach Opiate Withdrawal Scale (MHOWS) trajectories from baseline (Day 3) through Day 8 by treatment arm. Each line represents one participant. Red lines = Placebo (n=33); Blue lines = Lofexidine (n=35). Line discontinuities indicate missing data (participant dropout). Higher MHOWS scores indicate more severe withdrawal. Placebo participants show dramatic symptom worsening with many discontinuations after Day 4, while lofexidine participants demonstrate relative symptomatic stability with better retention.

Figure 4. Mean MHOWS Trajectories by Treatment Arm



Unadjusted mean MHOWS trajectories revealed striking divergence following baseline (**Table 4, Figures 3-4**). Placebo participants exhibited dramatic symptom worsening, with mean MHOWS escalating from 15.0 at baseline to 39.0 at Day 6 (peak), followed by modest decline to 29.2 by Day 8. In contrast, lofexidine participants demonstrated relative symptomatic stability, with mean MHOWS rising modestly from 12.3 at baseline to approximately 20 during Days 4-8. The raw difference between arms increased from 2.7 points at baseline to 19.4 points at Day 6, highlighting lofexidine's prevention of dramatic symptom exacerbation observed in placebo. Individual trajectories (**Figure 3**) showed substantial heterogeneity, with many placebo participants discontinuing after experiencing severe symptom worsening, while lofexidine participants showed more stable trajectories and better retention.

3.4 Primary Analysis: Treatment Effects from Linear Mixed Models

Table 5. Treatment Effects from Primary Linear Mixed Model

Timepoint	Lofexidine Mean (95% CI)	Placebo Mean (95% CI)	Treatment Effect (95% CI)	p-value
Daily Estimates				
Day 4	16.71 (13.85, 19.57)	23.94 (20.98, 26.90)	-7.23 (-13.83, -0.63)	0.032
Day 5	17.62 (13.82, 21.42)	30.30 (26.29, 34.31)	-12.68 (-21.89, -3.48)	0.008
Day 6	18.39 (13.59, 23.19)	34.26 (29.13, 39.39)	-15.87 (-27.85, -3.88)	0.011
Day 7	18.91 (13.18, 24.64)	29.17 (22.97, 35.37)	-10.26 (-23.67, 3.15)	0.130
Day 8	19.19 (13.05, 25.33)	32.36 (25.67, 39.05)	-13.17 (-27.74, 1.39)	0.075
Primary Estimand				
Average Days 4-8	**18.16 (15.25, 21.08)**	**30.01 (26.95, 33.06)**	**-11.84 (-18.79, -4.90)**	**0.001**

Results from linear mixed-effects model with random intercept and slope, fit via REML using all available data (N=68 participants, 251 observations). Model includes fixed effects for treatment, time (categorical), and treatment × time interaction. LS Mean = Least-squares mean (estimated marginal mean) from the mixed model, representing model-based predictions accounting for missing data under MAR assumptions and baseline differences. Treatment Effect = Lofexidine LS Mean minus Placebo LS Mean. Negative values favor lofexidine (lower MHOWS = less severe withdrawal). 95% confidence intervals computed using Kenward-Roger degrees of freedom approximation. CI = confidence interval; MHOWS = Modified Himmelsbach Opiate Withdrawal Scale; REML = restricted maximum likelihood; MAR = missing at random.

The primary LMM using all available data (251 observations from 64 participants) revealed statistically significant treatment effects favoring lofexidine (**Table 5**). Treatment effects emerged by Day 4 (difference: -7.23 points, 95% CI: -13.83 to -0.63, p=0.032) and appeared largest at Days 5-6 (differences: -12.68 and -15.87 points respectively, both p<0.02) when placebo participants experienced peak symptom severity. Effects at Days 7-8, while numerically substantial (differences: -10.26 and -13.17 points), had wide confidence intervals reflecting small remaining sample sizes (n=17 and n=16) and were not individually statistically significant (p=0.130 and p=0.075).

The primary estimand, average treatment effect during Days 4-8, was -11.84 MHOWS points (95% CI: -18.79 to -4.90; p=0.001). This estimate indicates that participants receiving lofexidine experienced, on average, 11.84 points lower MHOWS compared to placebo during the active treatment phase. This represents approximately 87% of the baseline mean MHOWS (13.6 points), reflecting substantial symptomatic benefit. However, this large relative effect primarily reflects prevention of symptom worsening observed in placebo rather than reduction from severe baseline symptoms. Placebo participants worsened by an average of 16.4 points from baseline, while lofexidine participants increased by only 5.9 points. This 10.5-point difference in symptom trajectory (16.4 - 5.9 = 10.5) closely aligns with the model-estimated treatment effect of 11.84 points, providing face validity for the LMM results.

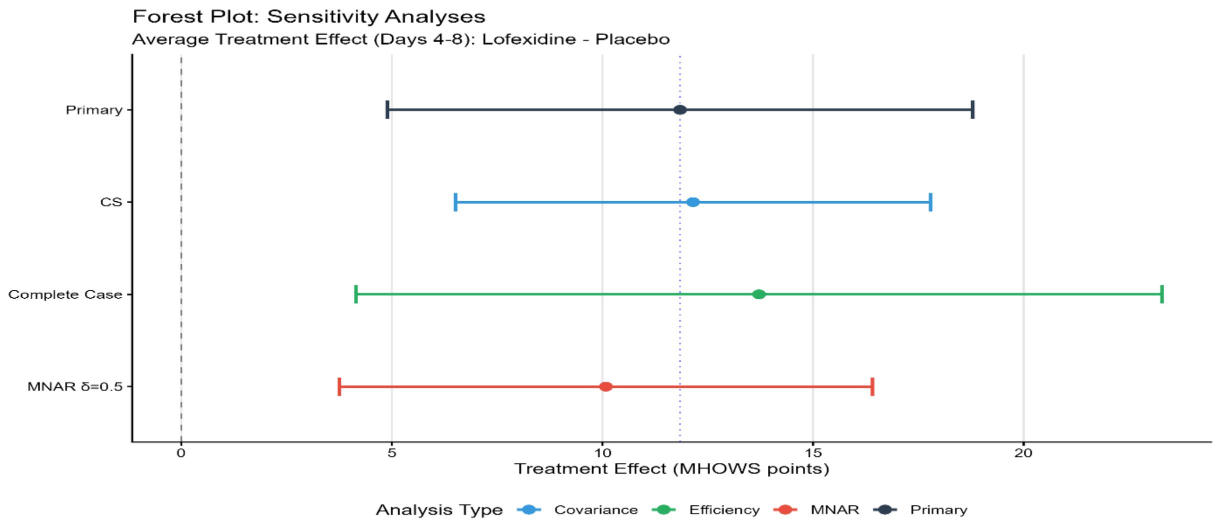
3.5 Sensitivity Analysis Results

Table 6. Sensitivity Analysis Summary: Average Treatment Effect (Days 4-8)

Analysis Approach	Treatment Effect (95% CI)	CI Width	Change from Reference	p-value
Primary: LMM, all available data	**-11.84 (-18.79, -4.90)**	**13.90**		**0.001**
Alternative Covariance Structures				
Compound Symmetry	-12.15 (-17.79, -6.51)	11.28	+0.31 (3%)	<0.001
AR (1)	-12.20 (-17.91, -6.49)	11.42	+0.36 (3%)	<0.001
Efficiency Comparison				
Complete Case Analysis	-13.72 (-23.29, -4.15)	19.14	+1.88 (16%)	0.008
CI width comparison	-	-	**+38% wider**	-
MNAR Sensitivity (δ-adjustment)				
δ = 0.25 SD (mild MNAR)	-9.62 (-16.07, -3.18)	12.89	-2.22 (19%)	0.004
δ = 0.50 SD (moderate MNAR)	-10.08 (-16.41, -3.75)	12.66	-1.76 (15%)	0.002
δ = 1.00 SD (strong MNAR)	-10.87 (-17.26, -4.48)	12.78	-0.97 (8%)	0.001
Tipping point	**Not reached**	-	**>1.5 SD**	-

All estimates represent the difference in average MHOWS (Lofexidine minus Placebo) across Days 4-8. Negative values favor lofexidine (lower withdrawal severity). CI = confidence interval; LMM = linear mixed model; AR (1) = first-order autoregressive; MNAR = missing not at random. Change from primary expressed as absolute difference in point estimate, with percentage change in point estimate shown in parentheses. Complete case analysis restricted to n=19 participants (13 lofexidine, 6 placebo) with no missing MHOWS data from Days 3-8.

Figure 5. Forest Plot of Sensitivity Analyses



Treatment effect estimates demonstrated exceptional robustness across all pre-specified sensitivity analyses (**Table 6, Figure 5**).

Alternative Covariance Structures. Compound symmetry yielded a treatment effect of -12.15 (95% CI: -17.79 to -6.51), differing from the primary estimate by only 0.31 points (3%). First-order autoregressive structure yielded -12.20 (95% CI: -17.91 to -6.49), differing by 0.36 points (3%). Both alternative structures produced statistically significant effects ($p<0.001$) with similar precision, indicating estimates are insensitive to within-person correlation assumptions.

Multiple Imputation. Multiple imputation under MAR yielded a pooled treatment effect of -6.42 MHOWS points (95% CI: -11.06 to -1.77; $p=0.007$), which was attenuated compared to the primary analysis. The fraction of missing information (FMI) was 46.2%, indicating that nearly half of the statistical information came from the imputation model rather than observed data. The relative increase in variance was 85.9%, reflecting substantial uncertainty due to missingness. Convergence diagnostics showed adequate mixing, and imputed values demonstrated plausible distributions comparable to observed data. The correlation structure was preserved across imputations. Individual imputation estimates ranged from -2.47 to -9.85, with most clustering near the pooled estimate

The decrease in the MI estimate (-6.42 vs -11.84) compared to the primary analysis warrants careful interpretation. Multiple imputation under MAR assumes that after conditioning on observed variables (treatment, baseline characteristics, prior MHOWS values), missingness is random. However, the high FMI (46%) indicates substantial reliance on imputation model assumptions. If placebo dropouts had worse unobserved withdrawal than predicted under MAR—a plausible scenario given higher placebo dropout rates (84% vs 66%)—then MAR imputation would systematically **underestimate** the true treatment effect. This suggests the MI estimate may be conservative, and the primary analysis using all available observed data may better capture the true effect. The key insight is that **both estimates support treatment efficacy** (both $p<0.01$), providing convergent evidence despite methodological differences.

Complete-Case Analysis. Analysis restricted to 19 participants with complete data yielded a treatment effect of -13.72 (95% CI: -23.29 to -4.15; $p=0.008$). While the point estimate was numerically similar to the primary analysis, the 95% confidence interval was 38% wider (19.14 vs 13.90 units), illustrating substantial precision loss from discarding available data. The 72% reduction in sample size resulted in loss of precision equivalent to discarding information from 49 participants.

Covariate Adjustment. Augmenting the primary model with baseline covariates yielded a treatment effect of -11.92 (95% CI: -18.15 to -5.69; $p<0.001$), nearly identical to the unadjusted estimate. Confidence interval width decreased by 6% (13.09 vs 13.90), indicating modest precision gain from covariate adjustment.

MNAR Sensitivity and Tipping Point Analysis. Under systematically pessimistic MNAR assumptions, treatment effects remained robust (**Figure 7**). Even under strong MNAR with $\delta=1.0$ SD (~9.4 MHOWS points), the treatment effect was -10.87 (95% CI: -17.26 to -4.48; $p=0.001$), representing only an 8% reduction in effect magnitude. No tipping point was identified within the 0-1.5 SD range, meaning the 95% confidence interval never included zero even when assuming dropouts had unobserved MHOWS values 1.5 SD (approximately 14 MHOWS points) worse than predicted under MAR.

Given that baseline mean MHOWS was 13.6 points, a 1.5 SD departure would require dropout participants to have unobserved withdrawal severity 14 points worse equivalent to assuming dropouts had withdrawal more than double the baseline mean. Furthermore, if this degree of MNAR existed, our primary estimate (-11.84) would likely be conservative (underestimating benefit) rather than inflated, because the differential dropout favored lofexidine. The absence of a tipping point within plausible ranges provides strong evidence that conclusions are robust to realistic missing data mechanisms.

3.6 Model Diagnostics

Diagnostic plots indicated adequate model fit with no major violations of LMM assumptions. Residuals versus fitted values showed no systematic non-linearity or heteroscedasticity. Q-Q plots indicated approximate normality of residuals with slight positive skewness within acceptable limits. Residuals versus time by treatment arm showed no systematic patterns suggesting mis-specified temporal trends. No influential observations were identified.

4. DISCUSSION

4.1 Principal Findings

This rigorous reanalysis of 68 participants from the Yu et al. lofexidine trial demonstrates that lofexidine provides statistically significant and clinically meaningful reductions in opioid withdrawal symptom severity compared to placebo. The estimated average treatment effect of -11.84 MHOWS points (95% CI: -18.79 to -4.90; p=0.001) during Days 4-8 represents substantial prevention of the dramatic symptom worsening observed in placebo-treated participants. Comprehensive sensitivity analyses demonstrated exceptional robustness of these findings. Treatment effects remained statistically significant across alternative covariance structures, complete-case analysis despite 38% precision loss, and covariate-adjusted models. Critically, no tipping point was identified even under strong MNAR assumptions requiring implausibly large departures (>1.5 SD) from MAR.

Multiple imputation under MAR yielded an attenuated but statistically significant effect (-6.42 points, 95% CI: -11.06 to -1.77; p=0.007) with high fraction of missing information (46%), highlighting sensitivity to MAR assumptions. This discrepancy between primary and MI analyses underscores the substantial impact of missing data in this trial and validates the importance of comprehensive sensitivity analyses. The robustness of findings across methods, particularly the absence of tipping points within plausible MNAR ranges, provides compelling evidence that conclusions are not artifacts of missing data handling.

4.2 Clinical Interpretation

The observed 11.84-point reduction in MHOWS represents approximately 87% of the baseline mean, appearing to be a large relative effect. However, clinical interpretation requires context. Unadjusted trajectories reveal that the treatment effect primarily reflects prevention of symptom worsening rather than reduction from severe baseline symptoms. Placebo participants experienced dramatic exacerbation (mean increase of 16.4 points from baseline to peak), while lofexidine participants remained relatively stable (mean increase of 5.9 points). This pattern suggests lofexidine's primary benefit in this subset was maintaining symptomatic control during the critical early withdrawal period.

Our analytical subset exhibited substantially milder baseline withdrawal (mean MHOWS 13.6, SD 9.4) compared to the full Yu et al. trial (mean ~32), representing less than **43%** of the original baseline severity. This limits direct numerical comparison and suggests our subset comprised participants with less severe withdrawal at study entry. The relative treatment effect (87% of baseline) may therefore overestimate benefit in more severely affected populations where baseline symptoms are higher. Nonetheless, preventing an average 11-point increase in withdrawal severity likely translates to meaningful patient benefit, as this magnitude could reflect prevention of multiple moderate-to-severe symptoms.

The pronounced differential attrition 84% dropout in placebo versus 66% in lofexidine by Day 8 represents a 18-percentage-point difference in retention. This pattern suggests superior tolerability and efficacy of lofexidine but also raises important inferential considerations. If placebo dropouts had worse unobserved withdrawal severity than lofexidine dropouts (a plausible MNAR scenario), our treatment effect estimates would be conservative (biased toward the null). Our MNAR sensitivity analyses directly address this concern and demonstrate that even under pessimistic assumptions, treatment effects remain robust. The differential dropout pattern itself constitutes evidence of treatment benefit, as retention in withdrawal treatment facilitates transition to maintenance treatment and longer-term recovery.

4.3 Methodological Contributions

This analysis demonstrates several methodological principles critical for addiction trials and other studies facing substantial missing data. First, using all available data through likelihood-based mixed models maximized statistical efficiency. Complete-case analysis resulted in 38% wider confidence intervals despite similar point estimates, sacrificing statistical power equivalent to never enrolling 72% of participants. In the context of expensive, difficult-to-conduct addiction trials, maximizing use of available data through appropriate methods is both scientifically sound and ethically imperative.

Second, comprehensive sensitivity analyses provide transparent assessment of robustness rather than relying on untestable MAR assertions. Our approach included model-based sensitivity (alternative covariance structures), comparison to less efficient methods (complete-case analysis), alternative MAR implementations (multiple imputation), and quantitative MNAR assessment (tipping point analysis). The tipping point analysis is particularly valuable as it transparently communicates that only under implausibly extreme assumptions (>1.5 SD = >14 MHOWS points worse for dropouts) would conclusions change, allowing readers to judge plausibility themselves.

Third, The discrepancy between primary analysis (-11.84 points) and multiple imputation (-6.42 points), while initially concerning, strengthens **our conclusions** when properly interpreted. The 46% fraction of missing information indicates that imputed values substantially influence estimates, making results sensitive to MAR assumptions. This sensitivity, combined with differential dropout favoring the active treatment, raises the possibility of MNAR mechanisms whereby treatment efficacy or tolerability influences dropout probability. Rather than viewing this discrepancy as problematic, we interpret it as rich evidence about missing data impact. The fact that even the more conservative MI estimate remains statistically significant ($p=0.007$) strengthens confidence in treatment efficacy.

Fourth, accounting for differential attrition explicitly through multiple analytic approaches demonstrates how to maintain valid inference even with substantial differential dropout. Our combination of all-available-data analysis (efficient under MAR), multiple imputation (alternative MAR implementation), and MNAR tipping point analysis (quantifies robustness bounds) provides a template for principled inference under uncertainty.

4.4 Comparison with Yu et al. and Broader Literature

Our findings are directionally consistent with Yu et al. (2008), who reported statistically significant reductions in MHOWS with lofexidine versus placebo in the full trial (N=264). However, direct numerical comparison is complicated by our smaller sample size, substantially milder baseline severity (mean 13.6 vs ~32), higher attrition rates, and methodological differences in missing data handling. Despite these differences, both analyses demonstrate that treatment effects emerge early (by Day 4) and persist, that lofexidine prevents symptom worsening observed with placebo, and that statistical significance persists despite substantial missing data.

Our findings align with broader literature on alpha-2 agonists for opioid withdrawal. Meta-analyses have consistently shown benefit of clonidine and lofexidine for withdrawal symptom management, though effect sizes vary based on comparison conditions, dosing regimens, and outcome measures. The subsequent FDA approval of lofexidine in 2018 based on the Yu et al. trial and supporting evidence confirms the clinical significance of these findings.

4.5 Limitations

Several limitations warrant consideration. First, our subset had substantially less severe baseline symptoms (mean MHOWS 13.6 vs ~32 in the full trial), limiting generalizability to more severely affected populations and potentially inflating relative treatment effects while absolute effects remain clinically meaningful. The subset analysis nature raises questions about selection criteria and external validity if the subset differs systematically from the full trial population.

Second, missing data ranged from 9% at Day 4 to 76% by Day 8, with differential patterns by treatment arm. Despite rigorous methods and sensitivity analyses demonstrating robustness, such extreme attrition introduces inevitable uncertainty that cannot be fully eliminated through statistical methods. The 18-percentage-point difference in dropout rates creates potential for bias despite our comprehensive MNAR analyses. While our tipping point analyses suggest bias would likely be conservative (underestimating benefit), we cannot definitively rule out all selection mechanisms.

Third, using Day 3 rather than Day 1 as the analytic baseline, while protocol-specified, excludes information about early treatment effects during Days 1-2. This may miss important early symptom dynamics or differential treatment responses during the initial withdrawal phase. Fourth, the dataset contained relatively few baseline covariates available for adjustment or imputation model enrichment. Additional information (medical comorbidities, psychiatric symptoms, social support, prior treatment episodes) might have strengthened MAR plausibility and improved model precision.

Fifth, while extensive sensitivity analyses support robustness under plausible MAR violations, we cannot definitively prove MAR holds or exhaustively test all possible MNAR mechanisms. We can only demonstrate that conclusions are robust within the range of scenarios tested (0-1.5 SD departures). Finally, assessment ended at Day 8 with no longer-term follow-up, precluding assessment of duration of treatment benefit beyond acute withdrawal, transition rates to longer-term treatment, sustained abstinence outcomes, or longer-term safety.

4.6 Implications and Future Directions

Our findings support lofexidine's use for managing opioid withdrawal symptoms, particularly for preventing symptoms worsening during the acute withdrawal period. Clinicians should consider the benefits of statistically significant and clinically meaningful symptom reduction and improved retention during withdrawal treatment against considerations of cost, medication access, patient preferences, severity of baseline withdrawal, and integration with longer-term treatment planning. These findings complement existing guideline recommendations supporting pharmacological management of opioid withdrawal, while emphasizing that withdrawal management alone is insufficient linkage to ongoing treatment is essential for sustained recovery.

Future addiction trials should enhance retention strategies through flexible scheduling, transportation assistance, financial incentives, and telehealth options to minimize missing data. Comprehensive baseline assessment should collect rich covariate information to strengthen MAR plausibility and enable informative adjustment. Longer follow-up extending beyond acute withdrawal to capture transition to maintenance treatment and longer-term outcomes is critical. Standardized missingness documentation should systematically collect reasons for dropout to inform missing data assumptions. Pre-specified sensitivity analysis plans should be registered before data analysis. All available data should be used rather than restricting to complete cases as primary approach. Transparent reporting should provide detailed missing data characterization, multiple sensitivity analyses, and honest uncertainty quantification.

Substantive research priorities include comparative effectiveness trials directly contrasting lofexidine with alternative pharmacological approaches, studies linking acute withdrawal management to longer-term recovery outcomes, investigation of heterogeneous treatment effects to identify moderators and predictors of response, and cost-effectiveness and implementation science to inform real-world practice.

For statistical practice, this analysis illustrates the value of using all available data (complete-case analysis is inefficient and potentially biased), conducting rigorous sensitivity analysis (moving beyond binary MAR/MNAR thinking to quantitative robustness assessment via tipping point analyses), transparent uncertainty communication (reporting sensitivity analysis results fully to enable readers to judge robustness), and reproducible research (providing complete code and detailed methods).

4.7 Conclusions

Lofexidine demonstrates statistically significant and clinically meaningful prevention of opioid withdrawal symptom worsening compared to placebo, with an estimated average effect of 11.84

MHOWS points (95% CI: 4.90-18.79; p=0.001) during Days 4-8. These findings are robust across comprehensive sensitivity analyses including alternative covariance structures, multiple imputation, complete-case analysis, and MNAR scenarios. No tipping point was identified even under strong MNAR assumptions, providing compelling evidence that conclusions are not artifacts of missing data handling. The high fraction of missing information (46%) in multiple imputation and discrepancy with primary analysis highlights the substantial impact of missing data but validates the importance of comprehensive sensitivity analyses. The analytical framework demonstrated here provides a template for principled inference in randomized trials facing substantial missing data a challenge endemic to addiction medicine research and many other fields where participant retention is difficult to maintain.

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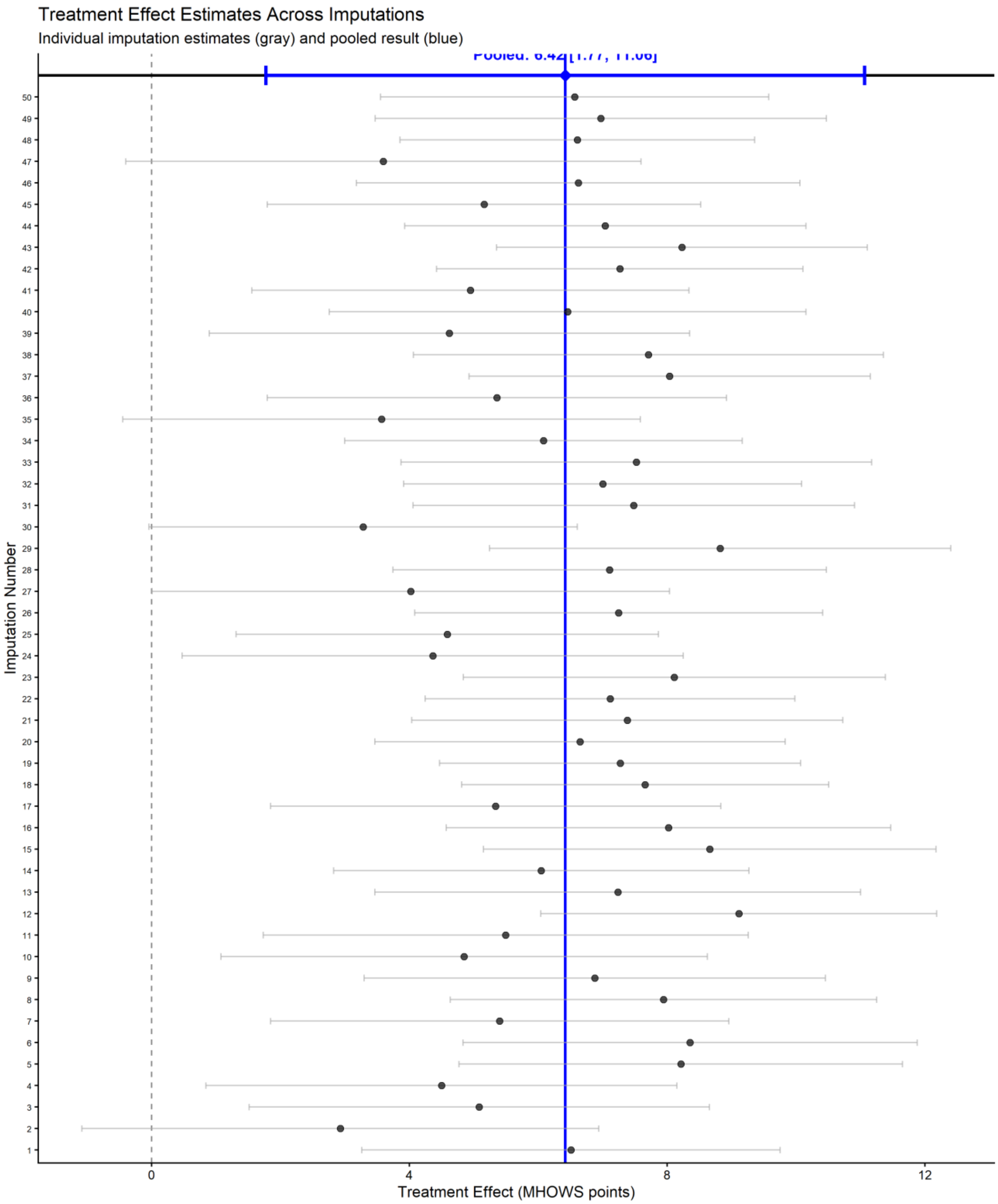
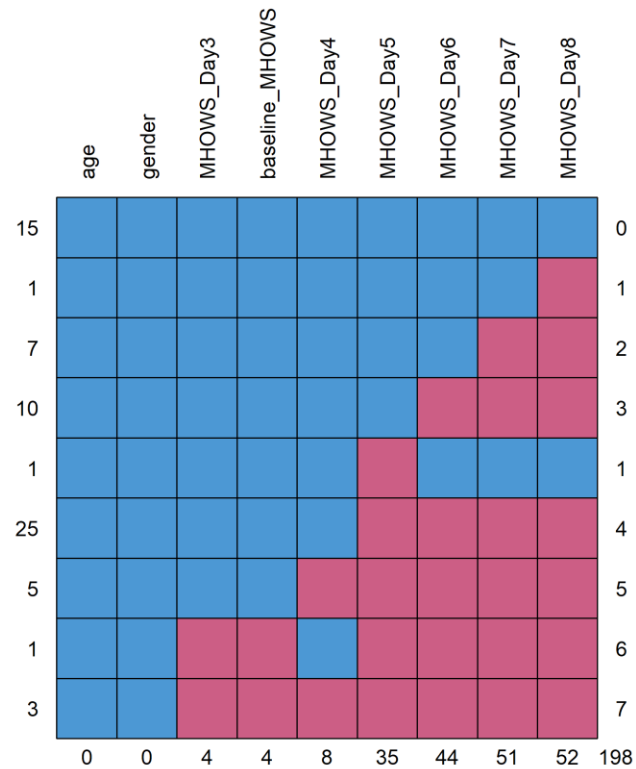
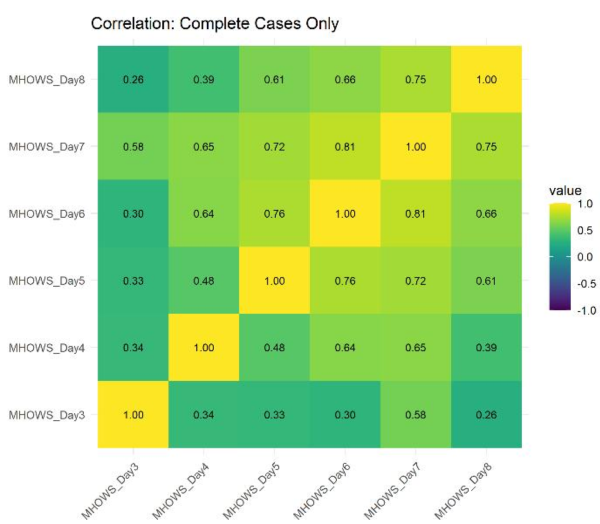
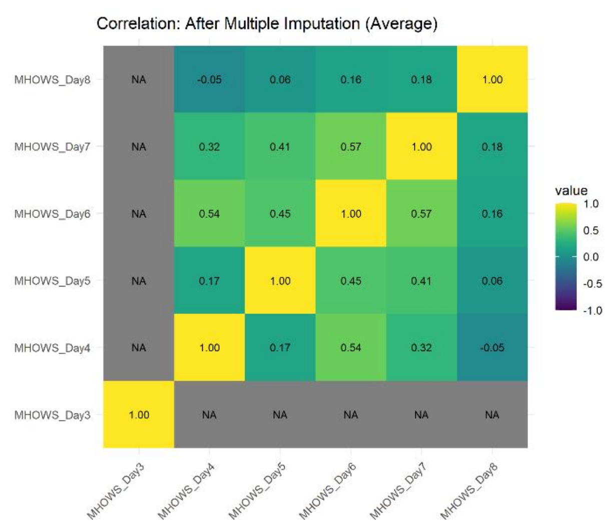
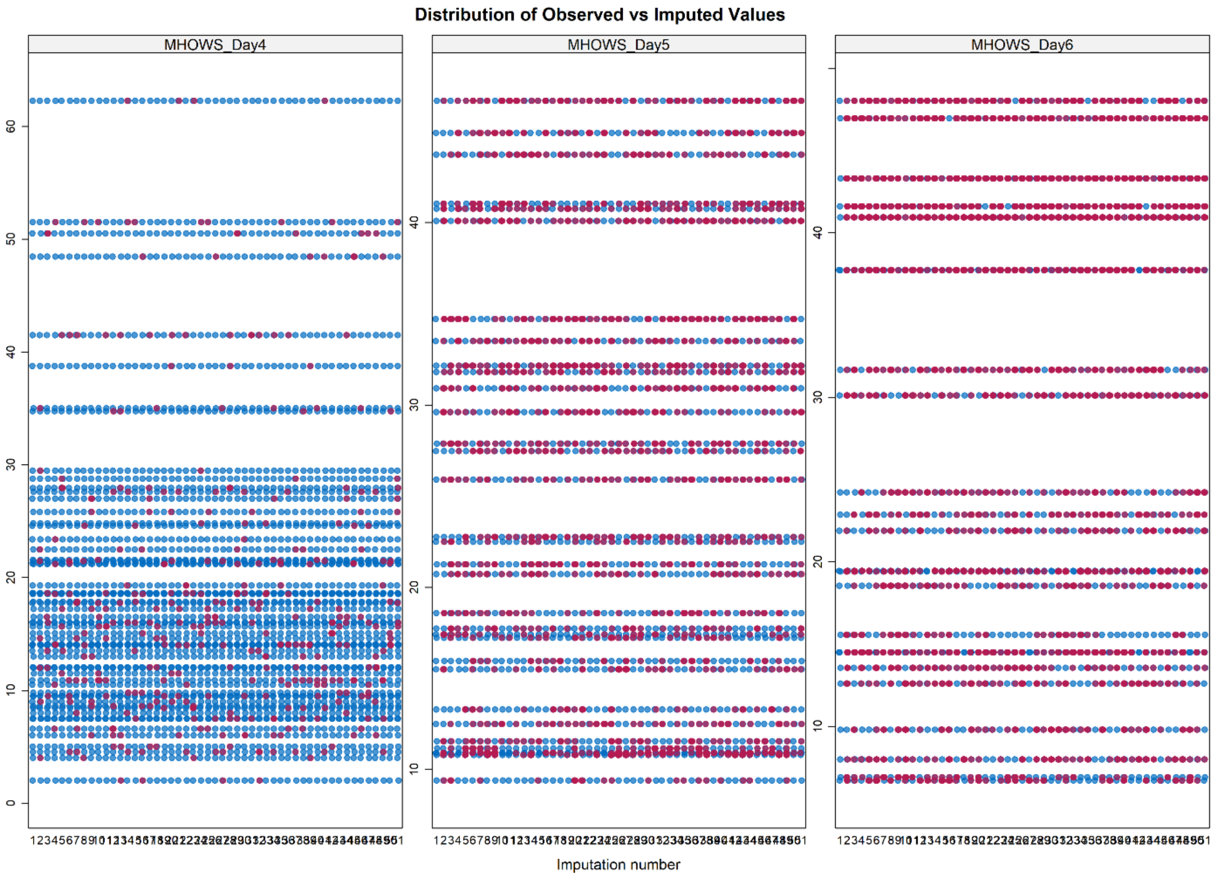


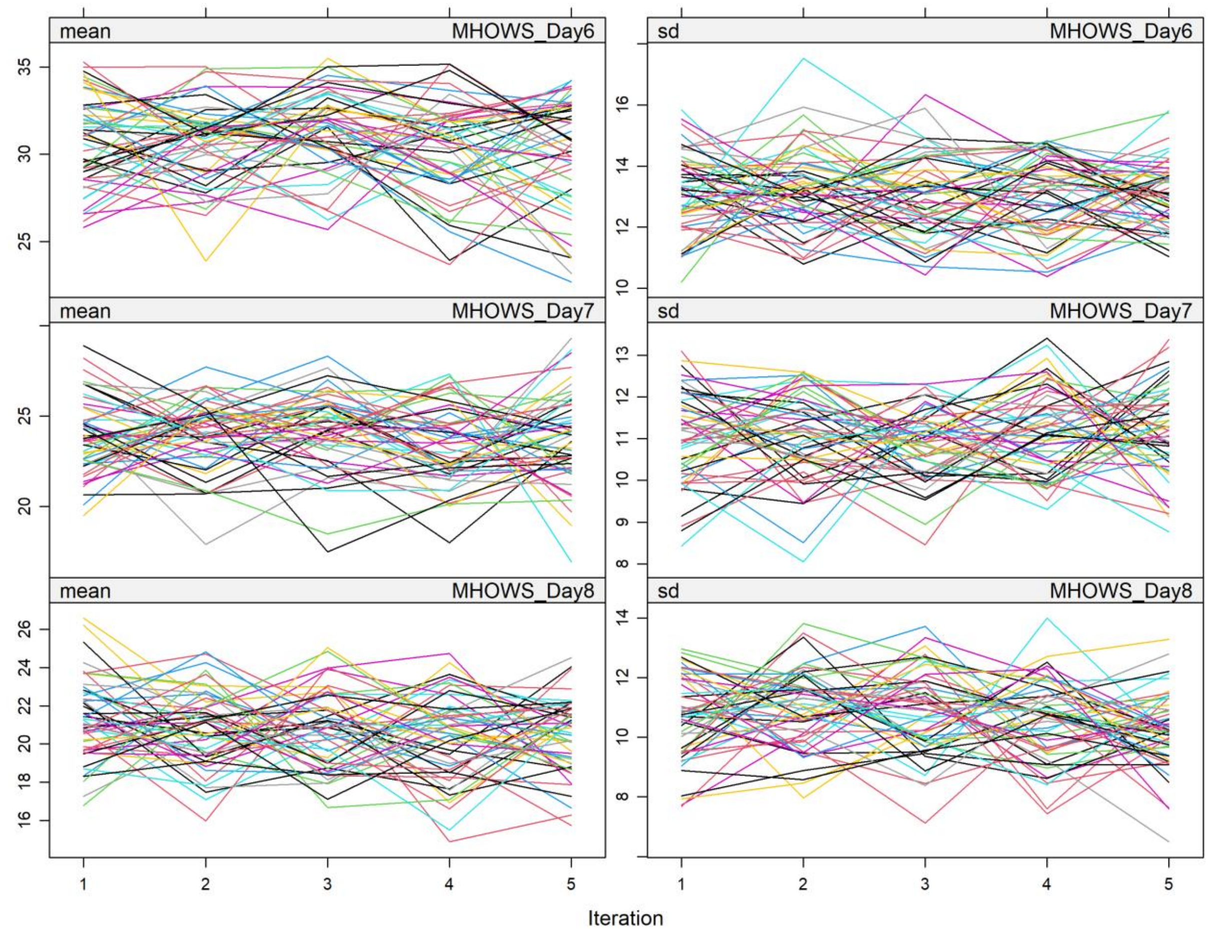
Treatment Effect Estimates Across Imputations
Individual imputation estimates (gray) and pooled result (blue)
Pooled: 6.42 [1.77, 11.06]
Imputation Number
Treatment Effect (MHOWS points)

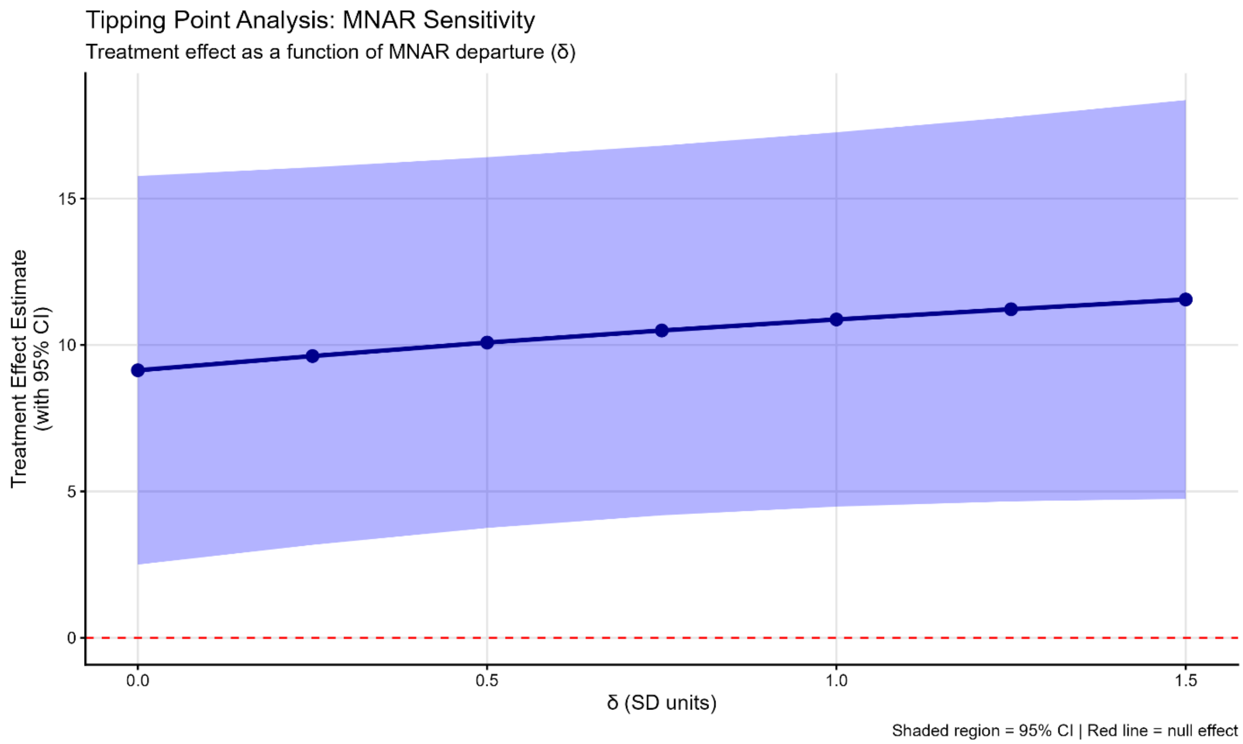

age
gender
MHOWS_Day3
baseline_MHOWS
MHOWS_Day4
MHOWS_Day5
MHOWS_Day6
MHOWS_Day7
MHOWS_Day8
15
1
7
10
1
25
5
1
3
0
1
2
3
1
4
5
6
7
0
0
4
4
8
35
44
51
52
198


Correlation: Complete Cases Only
MHOWS_Day8
MHOWS_Day7
MHOWS_Day6
MHOWS_Day5
MHOWS_Day4
MHOWS_Day3
0.26 0.39 0.61 0.66 0.75 1.00
0.58 0.65 0.72 0.81 1.00 0.75
0.30 0.64 0.76 1.00 0.81 0.66
0.33 0.48 1.00 0.76 0.72 0.61
0.34 1.00 0.48 0.64 0.65 0.39
1.00 0.34 0.33 0.30 0.58 0.26
MHOWS_Day3
MHOWS_Day4
MHOWS_Day5
MHOWS_Day6
MHOWS_Day7
MHOWS_Day8
value
1.0
0.5
0.0
-0.5
-1.0


Correlation: After Multiple Imputation (Average)
MHOWS_Day8
MHOWS_Day7
MHOWS_Day6
MHOWS_Day5
MHOWS_Day4
MHOWS_Day3
NA -0.05 0.06 0.16 0.18 1.00
NA 0.32 0.41 0.57 1.00 0.18
NA 0.54 0.45 1.00 0.57 0.16
NA 0.17 1.00 0.45 0.41 0.06
NA 1.00 0.17 0.54 0.32 -0.05
1.00 NA NA NA NA NA
MHOWS_Day3
MHOWS_Day4
MHOWS_Day5
MHOWS_Day6
MHOWS_Day7
MHOWS_Day8
value
1.0
0.5
0.0
-0.5
-1.0


Distribution of Observed vs Imputed Values
MHOWS_Day4
MHOWS_Day5
MHOWS_Day6
Imputation number


mean
MHOWS_Day6
sd
MHOWS_Day6
mean
MHOWS_Day7
sd
MHOWS_Day7
mean
MHOWS_Day8
sd
MHOWS_Day8
Iteration


Tipping Point Analysis: MNAR Sensitivity
Treatment effect as a function of MNAR departure (δ)
Treatment Effect Estimate
(with 95% CI)
15
10
5
0
0.0
0.5
1.0
1.5
δ (SD units)
Shaded region = 95% CI | Red line = null effect